

# Programming Methodology - SE1012

## Lab 2: Introduction to C Programming



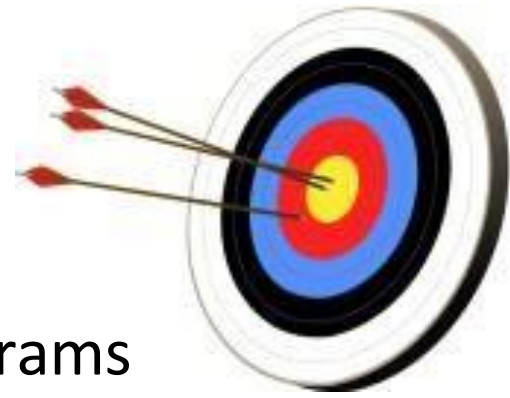
# Using Linux machines

**Access Lab 2 in GitHub via:**

<https://classroom.github.com/a/95CoYKqE>

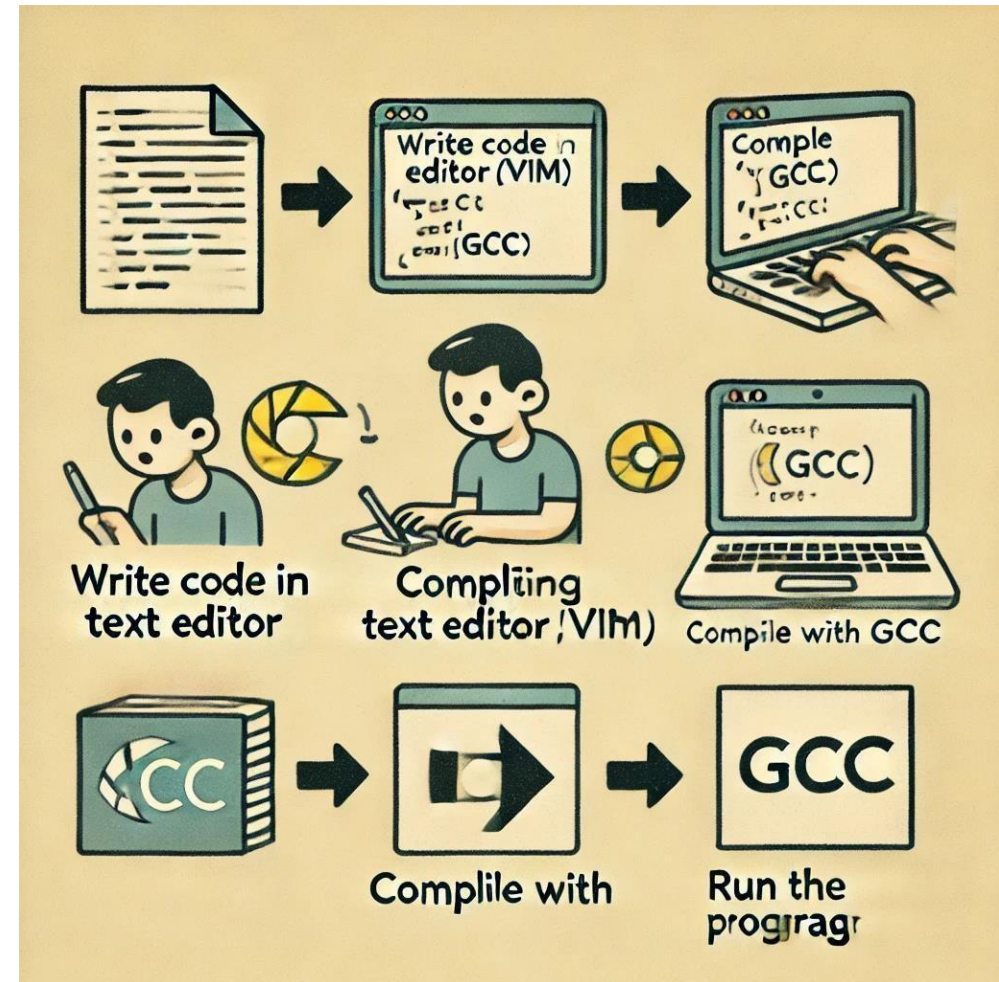
**Clone the Repository**

# Outline of Today's Lab



- Essential Commands for Creating, Compiling, and Running C Programs
- Activity 1: Displaying Text in C
- Activity 2: Calculating Missing Heights
- Activity 3: Calculating Fence Dimensions
- Lab Submission Guidelines Using GitHub - Command Line

# Essential Commands for Creating, Compiling, and Running C Programs



# Essential Commands

| Command          | Description                                         | Example                                         |
|------------------|-----------------------------------------------------|-------------------------------------------------|
| <code>vim</code> | Open the Vim text editor to create or edit files    | <code>vim hello_world.c</code>                  |
| <code>gcc</code> | GNU Compiler Collection, used to compile C programs | <code>gcc hello_world.c -o hello_world.o</code> |
| <code>./</code>  | Run the compiled C program                          | <code>./hello_world.o</code>                    |

# Essential Commands

1. Open the terminal and navigate to the directory where you want to create the file.
2. Type `vim hello_world.c` and press `Enter`.
3. In Vim, press `i` to switch to Insert mode, then type the C program.

```
#include <stdio.h>

int main() {
    printf("Hello, World!\n");
    return 0;
}
```

4. Press `Esc` to switch to Normal mode, then type `:wq` and press `Enter` to save and exit.
5. Compile the program with `gcc hello_world.c -o hello_world.o`.
6. Run the program with `./hello_world.o`.

# Other Ways for Compiling and Running Your Code

## 1. Without `.o` Extension:

```
gcc NameOfYourProgram.c -o NameOfYourProgram
```

- This command compiles `NameOfYourProgram.c` and produces an executable named `NameOfYourProgram`.
- To run the program, use:

```
./NameOfYourProgram
```

You do not need to add any extension to the executable name.

## 2. Compilation without `-o` Option (Not Recommended; Use only for simple programs):

```
gcc NameOfYourProgram.c
```

- This produces an executable named `a.out`.
- To run the executable:

```
./a.out
```

# Activity 1:

# Displaying Text in C

## Activity 1

Write a C program to display:

1. "Hello! Welcome to SLIIT"
2. "First line\nThis is the second line"

3. 

```
      x
    x  x
  x    x
x      x
x      x
x      x
x      x
x      x
      x
```

## Instructions

1. Open the Terminal using the Applications Menu or the keyboard shortcut (Ctrl + Alt + T).
2. Navigate to the Desktop.
3. Create a working directory with your IT number.
4. Inside the directory, create a new C file named `display_text.c`.
5. Write the C program as described in Problem 1.
6. Compile and run the program to ensure it works correctly.

# Activity 2: Calculating Missing Heights

In an experiment, the heights of 5 people were recorded and the average height was computed. Due to negligence, the height of two people has been lost. The lab assistant distinctly remembers that the two missing heights were almost the same. Develop a C program to input the heights of the known three people and the calculated average. Find and print the missing heights.

## Instructions

1. Open the Terminal using the Applications Menu or the keyboard shortcut (Ctrl + Alt + T).
2. Navigate to the working directory you created on the Desktop.
3. Create a new C file named `missing_heights.c`.
4. Write the C program as described in Problem 2.
5. Compile and run the program to ensure it works correctly.

# Activity 2: Calculating Missing Heights

```
c Copy code  
  
#include <stdio.h>  
  
int main() {  
    float height1, height2, height3, average, total, missing1, missing2;  
  
    printf("Enter the heights of three known people: ");  
    scanf("%f %f %f", &height1, &height2, &height3);  
  
    printf("Enter the average height: ");  
    scanf("%f", &average);  
}
```



***Complete code,  
Compile, and Run***

# Activity 3: Calculating Fence Dimensions

Write a C program to determine the width and length of a rectangular fence where the perimeter is known. If the width of the fence is  $\frac{3}{4}$  the size of the length of the fence. The program should take the perimeter as input and calculate the width and length.

## Instructions

1. Open the Terminal using the Applications Menu or the keyboard shortcut (Ctrl + Alt + T).
2. Navigate to the working directory you created on the Desktop.
3. Create a new C file named `fence_dimensions.c`.
4. Write the C program as described in Problem 3.
5. Compile and run the program to ensure it works correctly.

# Lab Submission Guidelines

- Compile a **small report including screenshots showing you work**
- **Rename the report** with **your IT number** and **covert it to a pdf** file (Example ITxxxxxxxxx.pdf)
- Upload the pdf file to the GitHub Classroom

